

Spatial Analysis of Electricity Access Inequality in Indonesia

Spatial Statistics and Machine Learning on District-Level INDO-DAPOER Data (2019)

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1 Introduction

1.1 Background

Indonesia is the world’s largest archipelagic state: more than 17,000 islands spread across three time zones, with enormous variation in geography, income, and infrastructure. Electricity is a foundational input for education, health services, and economic activity, and Indonesia has made remarkable progress in extending it: the national electrification programme pushed household access from roughly two-thirds of households in the 1990s to near-universal coverage today (World Bank, 2021; IEA, 2020).

That national success, however, hides a sharp spatial story. By 2019 the *median* Indonesian district (kabupaten/kota) had essentially closed its electricity gap, yet a small set of districts – concentrated in Papua, Nusa Tenggara Timur, and Maluku – still had a substantial share of households without electricity, in the most extreme case almost the entire district. The policy problem has therefore shifted from “raising the average” to “reaching the last places”, and those last places are not scattered at random: they cluster in specific corners of the archipelago.

This study examines that residual inequality using district-level data for 2019 from INDO-DAPOER, the Indonesia Database for Policy and Economic Research compiled jointly by the World Bank and BPS-Statistics Indonesia. The outcome of interest is the **electricity access deficit**: the percentage of households in a district *without* access to electricity.

1.2 Research Questions

1. How is the electricity access deficit distributed across Indonesian districts, and how large are the remaining disparities?
2. Which socio-economic characteristics are associated with the deficit?
3. Does the deficit exhibit spatial clustering, and where are the hotspots?
4. How well can machine learning models predict the district-level deficit from socio-economic variables, and which variables matter most?
5. What do simple policy scenarios suggest about the potential of complementary development interventions?

1.3 Significance

- **Academic**: adds district-level (rather than province-level) evidence on energy inequality in a developing archipelagic country, combining spatial econometrics with machine learning.
- **Practical**: identifies the specific district clusters where the remaining electrification effort should be concentrated.
- **Methodological**: demonstrates a fully reproducible pipeline from a public API to the final report; every figure in this document is generated from the raw data.

2 Literature Review

2.1 Energy Inequality

Energy inequality denotes disparities in access to, and quality of, modern energy services across social groups or regions. In developing countries it typically appears as an urban-rural or core-periphery divide, and it is increasingly measured not by national averages but by the distribution of access across sub-national units (IEA, 2020). When access approaches 100% nationally, the analytically interesting quantity becomes the *deficit* in the lagging tail of the distribution – the framing adopted here.

2.2 Regional Development Theory

Core-periphery models of regional development (Myrdal, 1957; Hirschman, 1958) predict that capital, labour, and infrastructure agglomerate in core regions, while peripheral regions face cumulative disadvantage: low density and difficult terrain raise the unit cost of infrastructure, low incomes weaken demand, and weak infrastructure in turn depresses income. Indonesia is a textbook case: Java-Bali forms the demographic and economic core, while the eastern islands – especially the Papuan highlands – are the classic periphery. This theory motivates two empirical expectations: (i) the access deficit should correlate with broader development indicators, and (ii) it should be spatially clustered rather than randomly scattered.

2.3 Determinants of Electricity Access

Prior empirical work points to a consistent set of correlates of electricity access in developing countries:

- **Income and expenditure:** richer households and richer regions can afford connections and attract grid investment.
- **Human development:** composite indices such as the HDI capture education and health conditions that co-evolve with infrastructure.
- **Poverty:** poverty headcounts flag the demand-side constraint directly.
- **Population density:** grid extension has strong economies of density; sparse settlement raises per-connection cost.
- **Complementary infrastructure:** access to safe water proxies the general reach of public service delivery into a district.

2.4 Spatial Methods and Machine Learning

Global Moran’s I (Moran, 1950) tests whether a variable is spatially autocorrelated overall; LISA statistics (Anselin, 1995) locate the specific clusters; and the Getis-Ord G_i^* statistic (Getis & Ord, 1992; Ord & Getis, 1995) identifies statistically significant hot- and coldspots. Spatial lag regression (Anselin, 1988) incorporates neighbourhood spillovers directly in a regression framework. On the predictive side, ensemble tree methods – Random Forest (Breiman, 2001) and gradient boosting (Chen & Guestrin, 2016) – and regularised regression (Friedman, Hastie & Tibshirani, 2010) are standard tools for tabular prediction and permit variable-importance and partial-dependence interpretation. This study combines the two strands.

3 Methodology

3.1 Data Sources

All data are original, public, and fully traceable; the raw API responses are archived in the project's `data/raw_dapoer/` folder and `build_dataset.R` re-downloads any missing file automatically from the original public APIs, so the dataset is both included with the submission and automatically downloadable in code. This dataset is not among the well-known teaching datasets (iris, titanic, MNIST, MovieLens, etc.) and this project is unrelated to the MovieLens recommender project.

1. **INDO-DAPOER** (Indonesia Database for Policy and Economic Research), World Bank & BPS-Statistics Indonesia. District-level indicators for **2019** – the most recent year with complete district coverage – retrieved from the World Bank API v2 (source id 45) on 14 July 2026.
2. **geoBoundaries** gbOpen release, Indonesia ADM2 boundaries (519 units, 2020 vintage; Runfola et al., 2020), used for district centroids and land areas.

The 514 DAPOER districts were matched to boundary polygons by normalised name and type (regency/city); nine districts required documented manual matches for official renames (e.g. Kabupaten Pontianak became Mempawah under Government Regulation 58/2014). Six districts are excluded for missing 2019 values: Buton Selatan, Buton Tengah, and Muna Barat (created in 2014, no population figures in DAPOER) and Deiyai, Mamberamo Tengah, and Puncak in Papua (no safe-water figures). This leaves **508 districts** with complete data on every variable used in the analysis.

```
# If the analysis outputs are missing, run the full pipeline once
# (the analysis script also rebuilds the dataset itself if needed).
if (!file.exists("output/final_analysis_results.csv")) {
  message("Analysis outputs not found -- running the analysis script now ...")
  source("analysis_electricity_access_inequality_indonesia_FINAL.R")
}
stopifnot(file.exists("output/final_analysis_results.csv"))
data <- read_csv("output/final_analysis_results.csv", show_col_types = FALSE)
cat("Districts loaded:", nrow(data), "\n")
```

```
## Districts loaded: 508
```

3.2 Variables

Variable	INDO-DAPOER series	Description
access_deficit	HOU.ELC.ACSN.ZS	100 minus % of households with electricity
hdi	IDX.HDI.REV	Human Development Index, revised method (0-100)
poverty_rate	SI.POV.NAPR.ZS	Poverty headcount (% of population)
expenditure_pc_thousand	HOU.XPD.PC.CR	Household expenditure per capita (thousand IDR/month)
grdp_per_capita	NA.GDP.INC.OG.SNA08.KRGRDP per person (constant-price, IDR million) / SP.POP.TOTL	
population_density	SP.POP.TOTL / polygon area	People per km ²
water_access	HOU.H2O.ACSN.ZS	% of households with safe water
literacy_rate	SE.LITR.15UP.ZS	Literacy, population 15+ (%)

Variable	INDO-DAPOER series	Description
<code>is_city</code>	location type	1 = kota (city), 0 = kabupaten (regency)

Density and GRDP per capita enter models in $\log_{10}$ form because both are heavily right-skewed.

3.3 Analytical Techniques

Exploratory analysis. Distributions, Spearman correlations (robust to the extreme skew of the deficit), and provincial rankings.

Spatial statistics. A k -nearest-neighbour spatial weights matrix ($k = 5$, row-standardised) built on district centroids – appropriate for an archipelago where contiguity is undefined across water. We compute global Moran’s I, LISA cluster maps, Getis-Ord G_i^* hotspots, and a spatial lag model (`spatialreg::lagsarlm`).

Machine learning. Four models – OLS, Random Forest, XGBoost, Elastic Net – trained on an 80% split with 10-fold cross-validation and compared on the 20% hold-out by RMSE, MAE, and R^2 . The best model is interpreted through variable importance and partial dependence.

Policy simulation. The best model predicts the national mean deficit under counterfactual scenarios: HDI +5 points, poverty –25%, safe-water access +10 percentage points, and all three combined. These are associative (not causal) what-if illustrations.

Software. All analyses were conducted in R (R Core Team, 2024). Spatial statistics use the `spdep` and `spatialreg` packages (Bivand et al., 2013), machine learning uses `randomForest` (Breiman, 2001; Liaw & Wiener, 2002), `xgboost` (Chen & Guestrin, 2016), and `glmnet` (Friedman et al., 2010), and graphics use `ggplot2` (Wickham, 2016).

4 Results and Discussion

4.1 Descriptive Statistics

```
desc <- data %>%
  select(access_deficit, hdi, poverty_rate, population_density,
         grdp_per_capita, water_access) %>%
  pivot_longer(everything(), names_to = "Variable") %>%
  group_by(Variable) %>%
  summarise(Min = min(value), Q1 = quantile(value, .25), Median = median(value),
            Mean = mean(value), Q3 = quantile(value, .75), Max = max(value),
            .groups = "drop")

show_table(desc, "Descriptive statistics, 508 districts (2019)")
```

Table 1: Descriptive statistics, 508 districts (2019)

Variable	Min	Q1	Median	Mean	Q3	Max
access_deficit	0.00	0.00	0.45	3.43	2.33	96.00
grdp_per_capita	0.01	18.70	26.77	38.17	39.92	487.90
hdi	30.75	66.44	69.23	69.69	72.84	86.65
population_density	0.83	54.25	151.51	1,118.61	814.43	20,539.55
poverty_rate	1.68	6.64	9.63	11.79	14.74	42.92
water_access	0.29	58.65	71.93	68.73	83.13	100.00

```
data %>%
  select(access_deficit, hdi, poverty_rate, log_pop_density) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = value)) +
  geom_histogram(aes(y = after_stat(density)), bins = 40,
                fill = "steelblue", alpha = 0.7) +
  geom_density(color = "darkred", linewidth = 0.8) +
  facet_wrap(~variable, scales = "free") +
  labs(x = NULL, y = "Density") +
  theme_minimal()
```

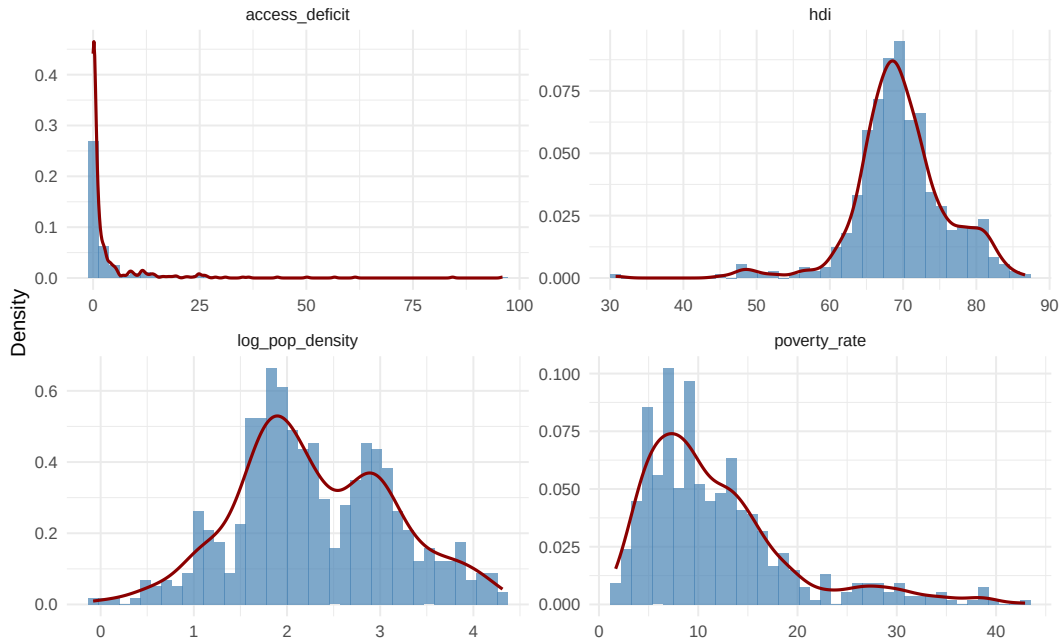


Figure 1: Distributions of the access deficit and key predictors

The most striking feature is the extreme skew of the outcome. The median district has a deficit of only 0.45%, and three-quarters of districts are below 2.33% – electrification is essentially complete for most of Indonesia. Yet the maximum reaches 96% (Kabupaten Dogiyai in the Papuan highlands, where only about 4% of households had electricity in 2019). The distribution of HDI is roughly symmetric around 69.7, while poverty and density are right-skewed, as expected.

4.2 Correlation Analysis

```
corr <- data %>%
  select(access_deficit, hdi, poverty_rate, expenditure_pc_thousand,
         log_pop_density, log_grdp_pc, water_access, literacy_rate) %>%
  cor(method = "spearman")

ggcorrplot(corr, hc.order = TRUE, type = "lower", lab = TRUE, lab_size = 2.8,
           colors = c("#B2182B", "white", "#2166AC"),
           ggtheme = theme_minimal)
```

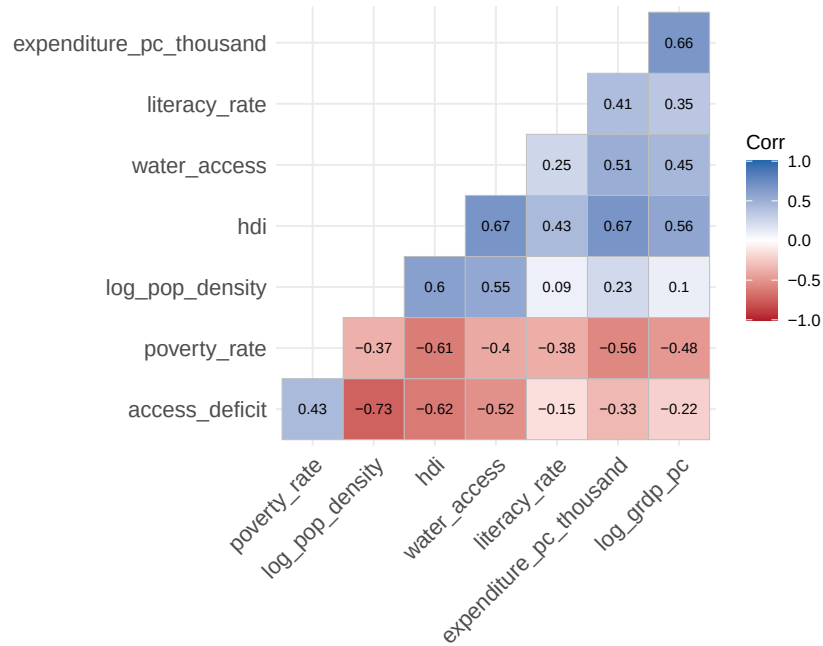



Figure 2: Spearman correlation matrix of key variables

Spearman correlations (used because of the outcome's skew) show the deficit is most strongly **negatively** associated with HDI ($\rho = -0.62$), literacy, per-capita expenditure, and safe-water access, and **positively** with the poverty rate ($\rho = 0.43$). The predictors are themselves inter-correlated – HDI, literacy, expenditure, and poverty form one development syndrome – which motivates the use of regularised and tree-based models later.

4.3 Provincial Disparities

```
prov <- read_csv("output/province_summary.csv", show_col_types = FALSE)

prov %>%
  ggplot(aes(x = reorder(province, mean_deficit), y = mean_deficit)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(x = NULL, y = "Mean share of households without electricity (%)") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 7.5))
```

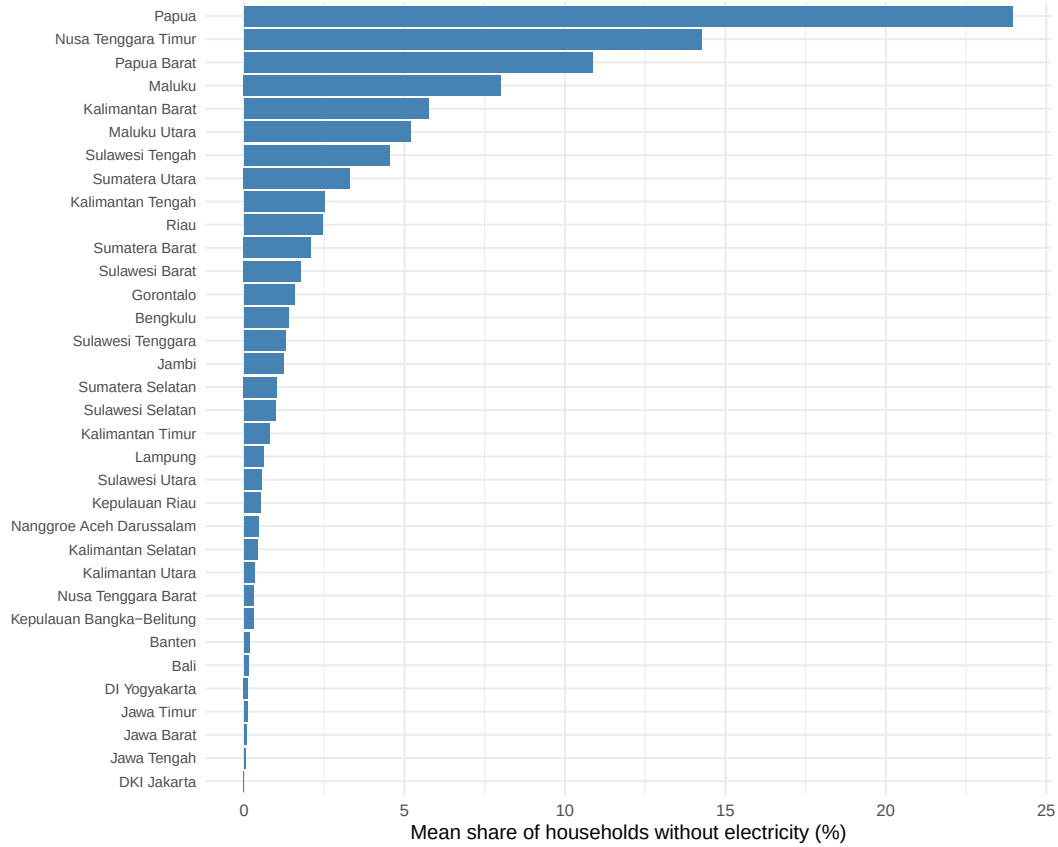


Figure 3: Mean electricity access deficit by province, 2019

```
prov_tbl <- bind_rows(
  prov %>% slice_max(mean_deficit, n = 5) %>% mutate(Group = "Highest deficit"),
  prov %>% slice_min(mean_deficit, n = 5) %>% mutate(Group = "Lowest deficit")
) %>%
  select(Group, Province = province, `Mean deficit (%)` = mean_deficit,
         `Worst district (%)` = worst_deficit, Districts = districts)

show_table(prov_tbl, "Provinces with the highest and lowest mean access deficit (2019)")
```

Table 2: Provinces with the highest and lowest mean access deficit (2019)

Group	Province	Mean deficit (%)	Worst district (%)	Districts
Highest deficit	Papua	23.96	96.00	26
Highest deficit	Nusa Tenggara Timur	14.26	36.47	22
Highest deficit	Papua Barat	10.86	43.21	13
Highest deficit	Maluku	8.01	25.01	11
Highest deficit	Kalimantan Barat	5.75	16.64	14
Lowest deficit	DKI Jakarta	0.00	0.00	6
Lowest deficit	Jawa Tengah	0.06	0.26	35
Lowest deficit	Jawa Barat	0.09	1.11	28

Lowest deficit	Jawa Timur	0.10	1.26	38
Lowest deficit	DI Yogyakarta	0.13	0.42	5

The provincial ranking is unambiguous. Papua's districts average a 24% deficit – an order of magnitude above the national norm – followed by Nusa Tenggara Timur (14.3%) and Papua Barat (10.9%). At the other end, the provinces of Java, Bali, and several Sumatran provinces are effectively at zero. Within provinces, heterogeneity can be large: Sumatera Barat averages only 2.1% yet contains Kepulauan Mentawai at roughly 26%, a reminder that isolated island districts lag even inside otherwise well-served provinces.

4.4 Spatial Analysis

4.4.1 Global Spatial Autocorrelation

```
coords <- as.matrix(data[, c("longitude", "latitude")])
nb <- knn2nb(knearneigh(coords, k = 5))
listw <- nb2listw(nb, style = "W")

moran <- moran.test(data$access_deficit, listw)
moran_tbl <- tibble(
  Statistic = c("Moran's I", "Expectation", "Variance", "p-value"),
  Value = c(round(moran$estimate[1], 4), round(moran$estimate[2], 4),
            round(moran$estimate[3], 6),
            signif(moran$p.value, 3))
)
show_table(moran_tbl, "Global Moran's I for the access deficit (k = 5 neighbours)",
           digits = 4)
```

Table 3: Global Moran's I for the access deficit (k = 5 neighbours)

Statistic	Value
Moran's I	0.4768
Expectation	-0.0020
Variance	0.0006
p-value	0.0000

Moran's I is 0.477 with a p-value below 10^{-15} : the access deficit is strongly and significantly spatially autocorrelated. Districts with large deficits neighbour other high-deficit districts.

4.4.2 Spatial Distribution and Hotspots

```
ggplot(data, aes(longitude, latitude, color = access_deficit)) +
  geom_point(size = 1.6, alpha = 0.85) +
  scale_color_viridis_c(option = "inferno", direction = -1) +
  labs(x = "Longitude", y = "Latitude", color = "Deficit (%)") +
  theme_minimal()
```

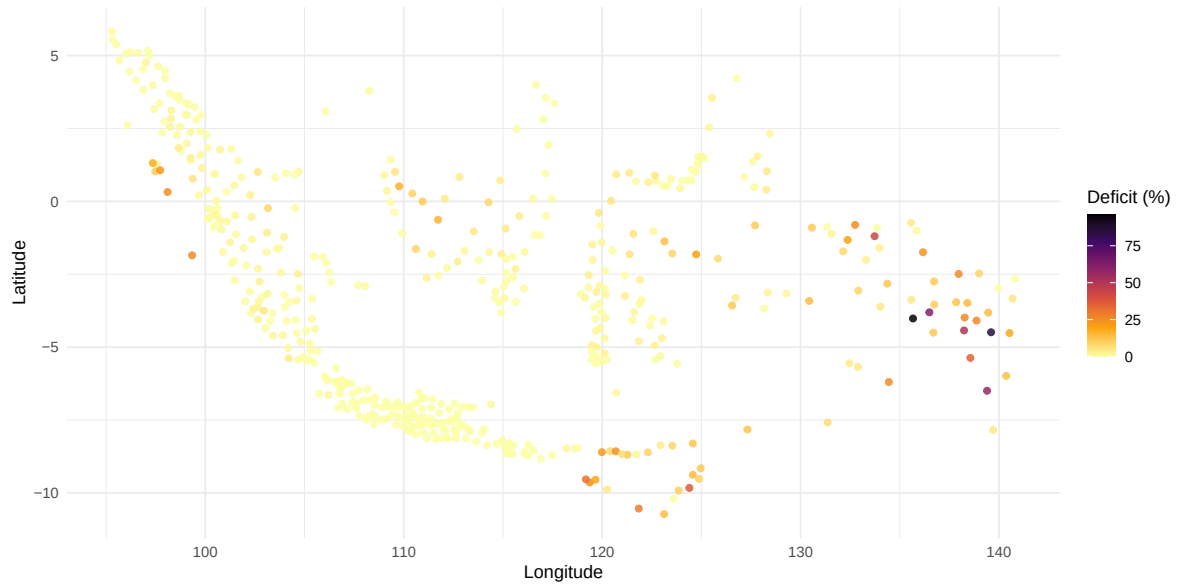


Figure 4: Access deficit at district centroids, 2019

```
ggplot(data, aes(longitude, latitude, color = lisa_cluster)) +
  geom_point(size = 1.6, alpha = 0.85) +
  scale_color_manual(values = c("High-High" = "#D73027", "Low-Low" = "#4575B4",
                                "High-Low" = "#FC8D59", "Low-High" = "#91BFDB",
                                "Not significant" = "grey80")) +
  labs(x = "Longitude", y = "Latitude", color = "Cluster") +
  theme_minimal()
```

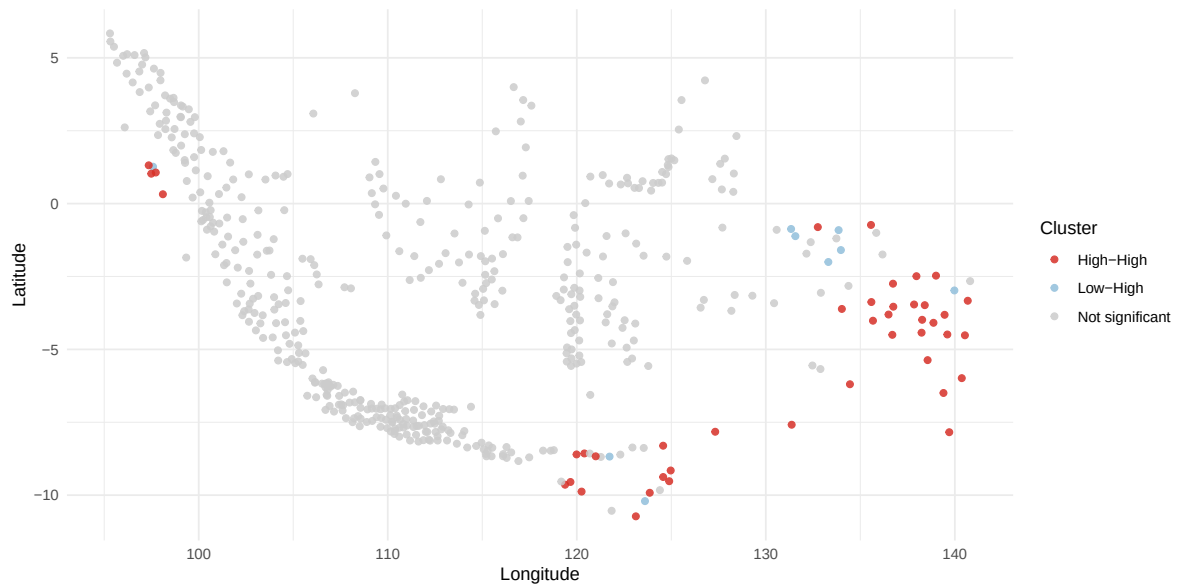


Figure 5: LISA clusters of the access deficit

```
ggplot(data, aes(longitude, latitude, color = hotspot)) +
  geom_point(size = 1.6, alpha = 0.85) +
  scale_color_manual(values = c("Hotspot" = "#D73027", "Coldspot" = "#4575B4",
                                "Not significant" = "grey80")) +
  labs(x = "Longitude", y = "Latitude", color = "Category") +
  theme_minimal()
```

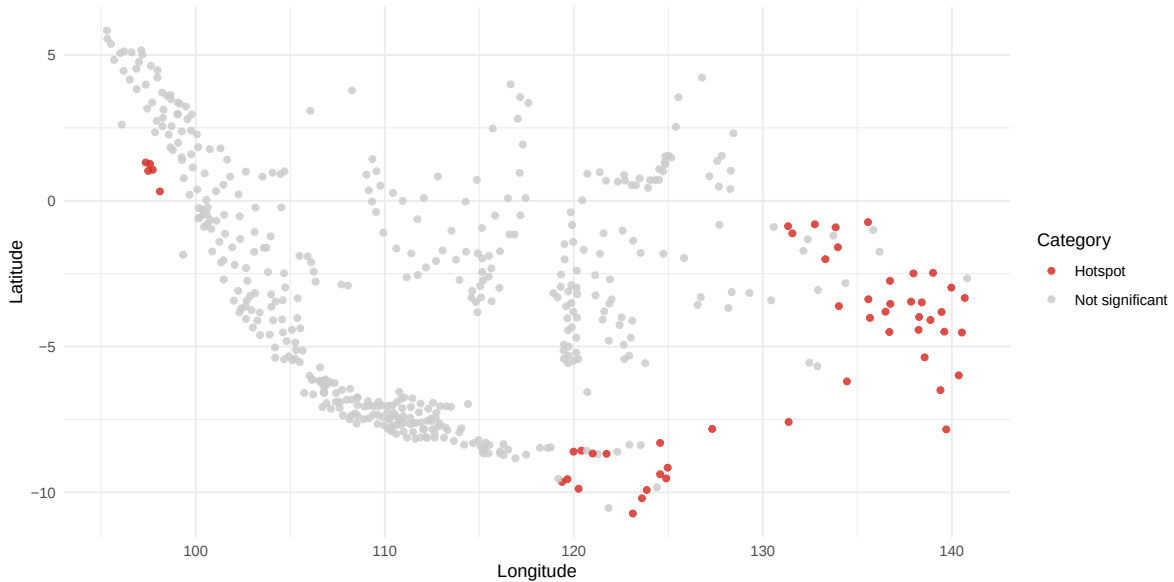


Figure 6: Getis-Ord G_i^* hotspots of the access deficit

```
cluster_tbl <- data %>% count(lisa_cluster, name = "Districts") %>%
  rename(`LISA cluster` = lisa_cluster)
show_table(cluster_tbl, "LISA cluster membership", digits = 0)
```

Table 4: LISA cluster membership

LISA cluster	Districts
High-High	43
Low-High	9
Not significant	456

The LISA analysis identifies 43 High-High districts – high deficits surrounded by high deficits – overwhelmingly in Papua (22) and Nusa Tenggara Timur (12), with the remainder in Maluku, Papua Barat, and the Nias islands of Sumatera Utara. The G_i^* statistic flags 52 hotspot districts in the same regions. There are no Low-Low clusters or coldspots: because most of the country sits at (near-)zero deficit, “low” is the norm rather than a distinctive cluster. Inequality in electricity access is thus a *one-tailed spatial phenomenon*, concentrated in a single eastern periphery, exactly as the core-periphery framework predicts.

4.4.3 Spatial Lag Regression

```

lag_model <- readRDS("output/spatial_lag_model.rds")
lag_sum <- summary(lag_model)

coef_tbl <- as.data.frame(lag_sum$Coef) %>%
  rownames_to_column("Term") %>%
  as_tibble() %>%
  mutate(across(where(is.numeric), ~ signif(.x, 3)))

show_table(coef_tbl, "Spatial lag model of the access deficit", digits = 3)

```

Table 5: Spatial lag model of the access deficit

Term	Estimate	Std. Error	z value	Pr(>#124;z#124;)
(Intercept)	32.300	5.920	5.460	0.000
hdi	-0.387	0.094	-4.110	0.000
poverty_rate	0.246	0.057	4.330	0.000
log_pop_density	-0.064	0.511	-0.126	0.900
log_grdp_pc	-0.794	0.941	-0.843	0.399
water_access	-0.080	0.021	-3.850	0.000
is_city	4.260	1.030	4.130	0.000

The spatial autoregressive parameter $\rho = 0.324$ is highly significant (LR test $p < 10^{-10}$): even after controlling for socio-economic covariates, a district's deficit rises with its neighbours' deficits, consistent with shared geography and shared grid infrastructure. Conditional on location, higher HDI and better water access predict smaller deficits, and higher poverty predicts larger ones. The positive city coefficient is a conditional effect – *given* its HDI and poverty level, a city has slightly more unserved households than a comparable regency – and should not be read as cities being worse off unconditionally. Log density and log GRDP per capita are not significant once the human-development variables and the spatial lag are included. The LM test for residual autocorrelation (statistic = 1.09, $p = 0.3$) indicates the lag term has absorbed the spatial dependence.

4.5 Predictive Modelling

4.5.1 Model Comparison

```

perf <- read_csv("output/model_performance.csv", show_col_types = FALSE)
show_table(perf, "Hold-out performance on the 20 percent test set",
  digits = 3)

```

Table 6: Hold-out performance on the 20 percent test set

Model	RMSE	MAE	R2
Random Forest	7.355	2.545	0.530
Linear	7.881	3.789	0.455
Elastic Net	7.933	3.729	0.449
XGBoost	8.032	2.549	0.448

Random Forest performs best, with hold-out RMSE 7.35 percentage points and R^2 0.53. The tree ensembles clearly beat the linear models on MAE (about 2.5 versus 3.8 points), indicating that they capture the many all-but-zero districts far more accurately; the RMSE gap is narrower because a handful of extreme Papuan districts dominate the squared error of every model. An honest reading is that socio-economic variables explain roughly half of the hold-out variance – geography, terrain, and infrastructure history, which are absent from the feature set, account for much of the rest.

4.5.2 Variable Importance

```
imp <- read_csv("output/variable_importance.csv", show_col_types = FALSE)
imp %>%
  ggplot(aes(x = reorder(variable, Overall), y = Overall)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(x = NULL, y = "Relative importance (0-100)") +
  theme_minimal()
```

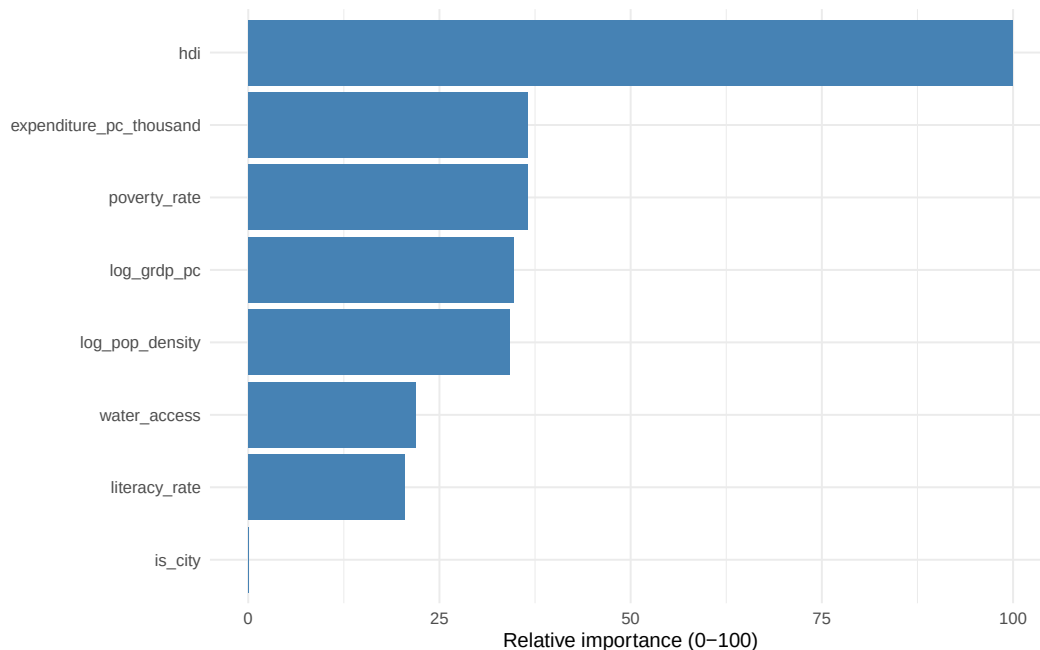


Figure 7: Variable importance of the best model

HDI dominates the importance ranking, at roughly three times the weight of the next group (per-capita expenditure, poverty, log GRDP per capita, log density). This mirrors the correlation and regression results: the electricity deficit is above all a *human-development* phenomenon.

4.5.3 Partial Dependence

```
include_graphics("output/partial_dependence.png")
```

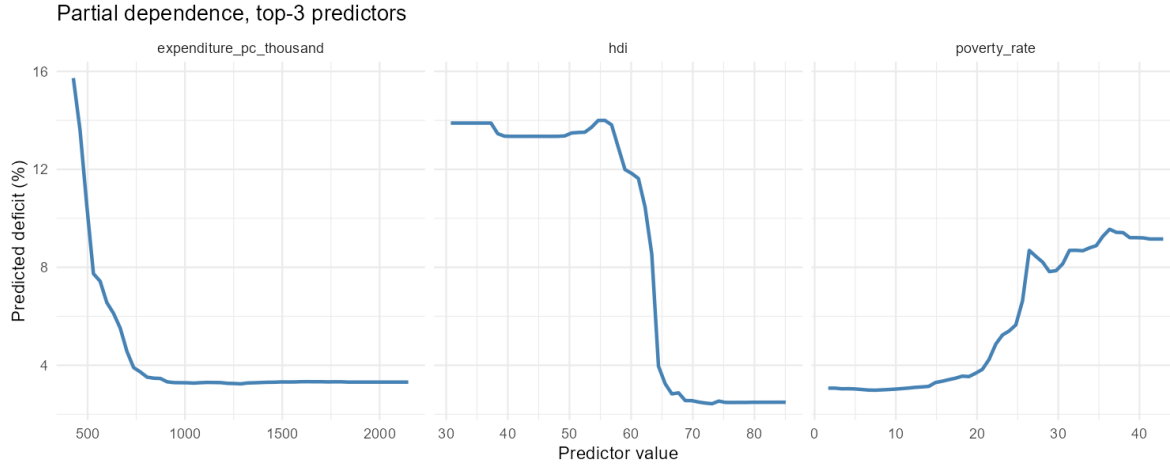


Figure 8: Partial dependence of the predicted deficit on the top-3 predictors

The partial dependence plots reveal a strongly non-linear HDI effect: the predicted deficit is flat and low for HDI above roughly 60, then rises steeply as HDI falls below that threshold. This threshold behaviour – invisible to a linear model – explains the tree models’ advantage and suggests that districts below HDI ≈ 60 constitute a qualitatively distinct policy group.

4.5.4 Policy Simulation

```
sim <- read_csv("output/policy_simulation.csv", show_col_types = FALSE)
show_table(sim %>%
  rename(`Mean predicted deficit (%)` = mean_deficit,
    `Change vs baseline (pp)` = change_vs_baseline),
  "Predicted national mean deficit under counterfactual scenarios",
  digits = 2)
```

Table 7: Predicted national mean deficit under counterfactual scenarios

Scenario	Mean predicted deficit (%)	Change vs baseline (pp)
Baseline	3.45	0.00
HDI +5 points	2.80	-0.65
Poverty -25%	3.16	-0.29
Water access +10 pp	3.21	-0.24
Combined	2.24	-1.21

```
sim %>%
  ggplot(aes(x = reorder(Scenario, -mean_deficit), y = mean_deficit)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = sprintf("%.2f", mean_deficit)), vjust = -0.4, size = 3.4) +
  labs(x = NULL, y = "Mean predicted deficit (%)") +
  theme_minimal()
```

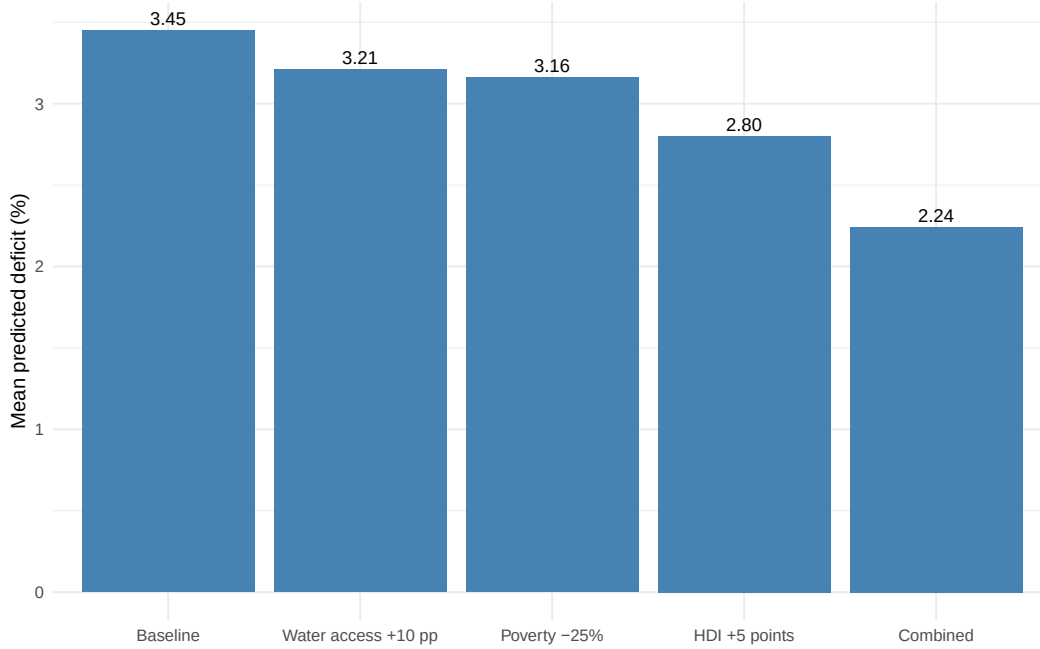



Figure 9: Scenario comparison of the predicted mean access deficit

Under the model, a five-point HDI improvement is associated with the largest single reduction in the predicted mean deficit (0.65 pp), and the combined scenario roughly a 1.2-pp reduction – about a third of the baseline mean. Two caveats are essential. First, these are *associative* counterfactuals from a predictive model, not causal estimates. Second, because the deficit is concentrated in a few dozen districts, national means understate what such changes would imply locally: the gains accrue almost entirely in the eastern hotspot districts.

4.6 Discussion

Three findings stand out. First, Indonesia’s electricity access problem in 2019 is no longer a broad national gap but a *residual, spatially concentrated* deficit: 43 High-High districts, essentially the Papuan highlands plus parts of Nusa Tenggara Timur and Maluku, against a near-universally electrified rest of the country. Second, the deficit is tightly bound to overall human development – HDI is the strongest correlate, regressor, and predictor throughout – with a threshold around HDI 60 below which the deficit escalates sharply. Third, spatial dependence survives socio-economic controls ($\rho \approx 0.32$), implying neighbourhood-level drivers (terrain, isolated grids, shared remoteness) beyond district characteristics.

For policy, this argues against uniform national programmes and for a targeted, spatially explicit strategy: off-grid and mini-grid solutions for the low-density highland hotspots, bundled with the education and poverty-reduction investments that the HDI linkage implies, rather than grid extension alone.

5 Conclusion and Recommendations

5.1 Summary of Findings

1. **Near-universal access with a hard tail.** The median district deficit is 0.45%, but Papua’s districts average about 24% and the worst district reaches 96%.
2. **Strong spatial clustering.** Moran’s $I = 0.48$ ($p < 10^{-15}$); all 43 High-High LISA districts and all 52 Gi* hotspots lie in eastern Indonesia; no coldspots exist because near-zero deficit is the national norm.
3. **Human development is the dominant correlate.** HDI, poverty, expenditure, and water access dwarf density and GRDP once compared jointly; the HDI effect is non-linear with a threshold near 60.
4. **Spatial spillovers persist.** The spatial lag model’s $\rho = 0.32$ remains significant after socio-economic controls.
5. **Models predict moderately well.** Random Forest attains hold-out $R^2 \approx 0.53$; the unexplained half plausibly reflects terrain and infrastructure history absent from the data.

5.2 Policy Implications

1. **Target the identified hotspots.** The 52 Gi* hotspot districts are a concrete priority list for the remaining electrification effort.
2. **Choose technology by geography.** Sparse highland and island districts are candidates for solar mini-grids and off-grid systems rather than costly grid extension.
3. **Bundle interventions.** The tight HDI-deficit link suggests electrification programmes should be coordinated with education, health, and poverty programmes in the same districts.
4. **Monitor at district level.** Provincial averages hide within-province pockets such as Kepulauan Mentawai; district-level (or finer) monitoring is necessary.

5.3 Limitations

1. **Cross-sectional design.** A single year (2019) precludes dynamic or causal analysis.
2. **Access is binary and supply-side.** The household-access measure ignores reliability, hours of supply, and affordability.
3. **Omitted geography.** Terrain ruggedness, distance to the existing grid, and island isolation are not in the feature set and likely explain much of the residual variance.
4. **Centroid-based weights.** k-NN weights on centroids approximate, but do not fully capture, real connectivity in an archipelago.
5. **Predictive, not causal.** All model-based scenario results are associations; they cannot support causal claims about interventions.

5.4 Future Research

1. Panel analysis of the 1996-2020 DAPOER series to study convergence in access.
2. Adding terrain, grid-distance, and night-lights covariates.
3. Causal evaluation of specific programmes (e.g. pre-electrification solar-lamp schemes) with quasi-experimental designs.
4. Measuring reliability and affordability, not just connection status.

5.5 Concluding Remark

Indonesia’s electrification achievement is real, but its final chapter is a spatial one. The remaining unserved households are concentrated in a small, well-identified set of eastern districts where low human develop-

ment, poverty, and geographic isolation reinforce one another. Data-driven spatial targeting – of the kind demonstrated here with fully public data – can help direct the last mile of the effort to exactly those places.

6 Acknowledgement of Sources and Academic Integrity

In line with the *Harvard Guide to Using Sources* (Harvard College Writing Program, n.d.) and the HarvardX honor code, the author confirms that this report is original work. All data are drawn from the publicly available INDO-DAPOER database (World Bank & BPS-Statistics Indonesia, 2022) and the geoBoundaries project (Runfola et al., 2020), both cited in full below with retrieval details. Every statistical method, theoretical framework, and software package used is attributed to its original authors through in-text citations and the reference list; all remaining analysis, interpretation, and writing are the author’s own. This work has not been submitted for credit in any other course.

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8 Appendix: Reproducibility

The complete pipeline is reproducible from public sources:

1. `build_dataset.R` – parses the archived INDO-DAPOER API responses (`data/raw_dapoer/`) and the geoBoundaries ADM2 file, matches districts, and writes `data/dapoer_district_2019.csv`.

2. analysis_electricity_access_inequality_indonesia_FINAL.R – runs the exploratory, spatial, and machine-learning analyses and writes all figures and tables to `output/`.
3. This document renders the report from those outputs.

sessionInfo()

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: Asia/Jakarta
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] spdep_1.4-2      sf_1.1-1          spData_2.3.5      ggcorrplot_0.2.0
## [5] kableExtra_1.4.1 knitr_1.51        lubridate_1.9.5    forcats_1.0.1
## [9] stringr_1.6.0    dplyr_1.2.1       purrr_1.2.2       readr_2.2.0
## [13] tidyr_1.3.2      tibble_3.3.1      ggplot2_4.0.3     tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1      viridisLite_0.4.3    farver_2.1.2
## [4] S7_0.2.2              fastmap_1.2.0        TH.data_1.1-5
## [7] digest_0.6.39         timechange_0.4.0     lifecycle_1.0.5
## [10] LearnBayes_2.15.2     survival_3.8-6       magrittr_2.0.5
## [13] compiler_4.6.1        rlang_1.3.0          tools_4.6.1
## [16] igraph_2.3.3          yaml_2.3.12          data.table_1.18.4
## [19] labeling_0.4.3        bit_4.6.0            sp_2.2-1
## [22] classInt_0.4-11       plyr_1.8.9           xml2_1.6.0
## [25] RColorBrewer_1.1-3    multcomp_1.4-30      KernSmooth_2.23-26
## [28] withr_3.0.3           grid_4.6.1           e1071_1.7-17
## [31] MASS_7.3-65           scales_1.4.0         cli_3.6.6
## [34] mvtnorm_1.4-1         rmarkdown_2.31       crayon_1.5.3
## [37] generics_0.1.4        otel_0.2.0           rstudioapi_0.19.0
## [40] reshape2_1.4.5        tzdb_0.5.0           DBI_1.3.0
## [43] proxy_0.4-29          splines_4.6.1        spatialreg_1.4-3
## [46] parallel_4.6.1        s2_1.1.11            marginaleffects_0.32.0
## [49] vctr_0.7.3           boot_1.3-32          Matrix_1.7-5
## [52] sandwich_3.1-1        hms_1.1.4            bit64_4.8.2
## [55] systemfonts_1.3.2     units_1.0-1          glue_1.8.1
## [58] codetools_0.2-20      stringi_1.8.7        gtable_0.3.6
## [61] deldir_2.0-4          pillar_1.11.1        htmltools_0.5.9
```

## [64]	R6_2.6.1	wk_0.9.5	textshaping_1.0.5
## [67]	vroom_1.7.1	evaluate_1.0.5	lattice_0.22-9
## [70]	backports_1.5.1	class_7.3-23	Rcpp_1.1.2
## [73]	svglite_2.2.2	coda_0.19-4.1	nlme_3.1-169
## [76]	xfun_0.60	zoo_1.8-15	pkgconfig_2.0.3